

Facility: SALEM 1 & 2 Scenario No.: ESG-1 Op-Test No.: 13-01 NRC

Examiners: _____ Operators: _____

Initial Conditions: 90% power, EOL.

Turnover: Raise Rx power to 98.5% @ 10% per hour IAW IOP-4, Power Operation.

Event No.	Malf. No.	Event Type*	Event Description
1		R CRS/RO N PO	Raise power
2	AN0266	C CRS	EDG Air Receiver low pressure (TS)
3	CC0172C CC0361B	C CRS/PO	CCW pump trip w / failure of standby pump to start (TS)
4	RC007D	C CRS/RO	RCP seal degradation
5	MS0092H	M ALL	Steam leak in containment
6	MS0092 E-H	C ALL	MSLI failure (Loss of Secondary Coolant)
7	RP318L1 RP318L2	C CRS/RO	Containment Spray pumps fail to start on SEC signal
			CT's: 1. Stop RCP with seal failure 2. Start minimum complement of containment cooling. 2. Minimize AFW flow

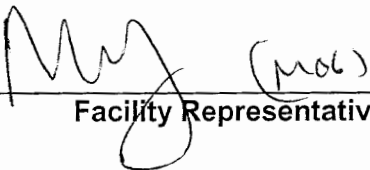
* (N)ormal, (R)eactivity, (I)nstrument, (C)omponent, (M)ajor

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 13-01 NRC ESG-1
SCENARIO NUMBER: 13-01 NRC ESG-1
EFFECTIVE DATE: See Approval Dates below
EXPECTED DURATION: 80 minutes
REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

Revision Summary
New Issue for 13-01 NRC Exam

PREPARED BY:	<u>G Gauding</u> Lead Regulatory Exam Author	<u>08-7-14</u> Date
APPROVED BY:	<u></u> Operations Training Manager	<u>10-23-14</u> Date
APPROVED BY:	<u></u> Facility Representative	<u>10-23-14</u> Date

I. OBJECTIVES

- A. Given the unit at >20% reactor power with the generator synchronized to the grid, the crew will perform a power increase at 10%/hr, IAW S2.OP-IO.ZZ-0004(Q).
- B. Given a situation with an OHA alarm, the crew will perform action(s) to investigate and correct the cause of the alarm, IAW approved station procedures.
- C. Given indication of a loss or malfunction of a safety related plant cooling water system, DIRECT the response to the loss or malfunction in accordance with the approved station procedures.
- D. Given the order or indications of a loss or malfunction of a safety related plant cooling water system, perform actions as the nuclear control operator to RESPOND to the loss or malfunction of the safety related plant cooling water system in accordance with approved station procedures.
- E. Given a RCP with various seal failures, take corrective action for RCP seal failures IAW AB.RCP-0001 and the appropriate alarm response procedures
- F. Given a failure of an RCP, perform actions as the nuclear control operator to RESPOND to the malfunction IAW approved station procedures.
- G. Given the failure of a RCP, DIRECT the response to the malfunction IAW approved station procedures.
- H. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with the approved station procedures.
- I. Given the order or indications of a reactor trip perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with the approved station procedures.
- J. Given indication of a safety injection DIRECT the response to the safety injection in accordance with the approved station procedures.
- K. Given the order or indications of a safety injection perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- L. Given a safety injection, START equipment that has failed to automatically start in accordance with the approved station procedures.
- M. Given the order or indications of a multiple steam generator depressurization, perform actions as the nuclear control operator to RESPOND to the generator depressurization in accordance with EOP-LOSC-2.
- N. Given indication of a multiple steam generator depressurization, DIRECT the response to the generator depressurization in accordance with EOP-LOSC-2
- O. Given indication of excessive containment pressure DIRECT the response to the excessive pressure in accordance with the approved station procedures.
- R. Given the order or indications of excessive containment pressure perform actions as the nuclear control operator to RESPOND to the excessive pressure in accordance with the approved station procedures.
- S. Given the unit in the EOPs and an excessive containment pressure the crew will reduce containment pressure IAW approved station procedures.

II. MAJOR EVENTS

- A. Power ascension
- B. 2C EDG air leak
- C. 23 CCW pump trip w/ failure of standby pump to auto start.
- D. 24 RCP seal degradation leading to failure.
- E. Steam rupture in containment w/ failure of MSIVs to shut.
- F. Containment Spray pump start failure.

III. SCENARIO SUMMARY

- A. The crew will take the turnover at 90% power, MOL, with direction to raise power to 98.5% @ 10% per hour. Rx Engineering is bringing updated reactivity plan to control room and crew should determine its own reactivity plan to establish ARO prior to reaching 98.5% power.
- B. After the power ascension is commenced, the crew will receive an urgent trouble alarm for 2C EDG. The CRS will determine that 2C EDG is inoperable based on lower than required Air Start Tank pressure, and enter the appropriate Tech Spec.
- C. Once the EDG Tech Spec is identified, 23 Component Cooling Water pump will trip. The standby CCW pump will not auto start. The crew will manually start the standby CCW pump to restore system pressure and clear CCW and RCP system alarms. The CRS will enter the appropriate Tech Spec.
- D. 24 RCP #1 seal will exhibit signs of degrading performance. The crew will enter S2.OP-AB.RCP-0001, Reactor Coolant Pump Abnormality, to address the rising seal leakoff flow. The crew will initiate a Rx trip when seal leakoff reaches the point where the RCP must be tripped. The crew will trip 24 RCP after the Rx is tripped.
- E. 3-5 minutes after the 24 RCP is tripped, the crew will isolate the seal leakoff return valve for 24 RCP, to prevent further damage and seal degradation of the 24 RCP. (CT #1).
- F. After performing Rx trip immediate actions in EOP-TRIP-1, Rx Trip or Safety Injection, the crew will transition to EOP-TRIP-2, Reactor Trip Response, when it is determined no condition exists which warrants a Safety Injection. CFST's will be active upon transition out of TRIP-1. During performance of RCS cooldown minimization steps, a steam leak will develop in containment. The crew will initiate a MSLI, then initiate a Safety Injection when all MSIVs remain open (MSLI fails).
- G. The crew will return to EOP-TRIP-1, then transition out to EOP-LOSC-1, Loss of Secondary Coolant. The crew will transition to EOP-LOSC-2, Multiple Steam Generator Depressurization due to all MSIVs being open.
- H. The crew will transition to EOP-FRCE-1 when containment pressure rises to 15 psig. The crew will diagnose no Containment Spray pumps are running and start Containment Spray pumps. (CT #3).
- I. During performance of LOSC-2 or FRCE-1 Response to Excessive Containment Pressure, the crew will reduce AFW flow to minimize the RCS cooldown. (CT#2).
- J. A single MSIV will be closed in the field during performance of LOSC-2. The crew will recognize rising pressure in the associated SG, and transition to LOSC-1.
- K. The scenario will terminate after the return to LOSC-1.

IV. INITIAL CONDITIONS

____ Pre-snapped IC-231

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
____ 1	2VC1 and 2VC4 C/T
____ 2	RCPs (SELF CHECK)
____ 3	RTBs (SELF CHECK)
____ 4	MS167s (SELF CHECK)
____ 5	500 KV SWYD (SELF CHECK)
____ 6	SGFP Trip (SELF CHECK)
____ 7	23 CV PP (SELF CHECK)
____ 8	S2.OP-IO.ZZ-0004, Power Operation, in progress and complete up to Step 5.1.26.
____ 9	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:		
Initial	ET #	Description
	3	EVENT ACTION: kb528ttx // 21 SG FEED PUMP TURB-TRIP COMMAND: PURPOSE: <update as needed>
	5	EVENT ACTION: kc201pb3 // 2C DIESEL GEN-START COMMAND: PURPOSE: <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	AN0266 SER 266 FAILS - :J20 2C DIESEL GENERATOR URGENT TROUBLE	N/A	N/A	N/A	RT-1	SER POINT FAILS/OVRD TO ON
02	RP318Q3 2C Diesel GEN Fails to Start on SEC	N/A	N/A	N/A	N/A	
03	CC0172C 23 COMPONENT COOLING PUMP TRIP	N/A	N/A	N/A	RT-3	
04	CC0361B 22 COMPONENT COOLING PUMP Fails to Start on Low Pressure	N/A	N/A	N/A	N/A	
05	RC007D 24 RC PUMP - #1 SEAL FAILS - w/ DEGRADATION	N/A	0.3	00:05:00	RT-3	0.75
06	MS0092E 21MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
07	MS0092F 22MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
08	MS0092G 23MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
09	MS0092H 24MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
10	MS0090Ar 21 Main Steam Line Leak Inside Cont AFTER orifice	00:01:00	6	00:05:00	ET-3	10
11	RP318L1 21 CS Pump Fails to Start on SEC	N/A	N/A	N/A	N/A	
12	RP318L2 22 CS Pump Fails to Start on SEC	N/A	N/A	N/A	N/A	
13	EL0163 2C EMERG DIESEL GENERATOR TRIP	00:00:10	N/A	N/A	RT-10	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
01	DG27D 2C DIESEL ENGINE LOCAL START	N/A	N/A	N/A	RT-10	START

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
01	C201 F DI 2C DIESEL GEN-STOP	00:00:05	N/A	N/A	ET-5	ON
02	C201 C1 LO 2C DIESEL GEN-READY TO LOAD	N/A	N/A	N/A	N/A	OFF
03	C203 L AO 2C DIESEL GENERATOR - VOLTS	N/A	N/A	N/A	N/A	0
04	C203 R AO 2C DIESEL GENERATOR - FREQ	N/A	N/A	N/A	N/A	35

OTHER CONDITIONS:

Description

1. None

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Power ascension Note: CRS may direct rod movement only until rods are fully withdrawn. If this is the method chosen, then dilution will not be performed prior to initiating the up power. Note: The following reactivity calculations should be used as a baseline to compare to the reactivity calculation performed by the candidates. Using the Graphs vs. Tables could result in a different reactivity change value being calculated. Note: Power defect change from 90% to 98.5% is 166.6 pcm. (Page 104/115 of S2.RE-RA.ZZ-0016 Curve Book, using 6,000 EFPH and 900 Cb RCS boron)) There is 58 pcm left in rods at 203 steps. (page 32/115). This means 108.6			
	CRS briefs crew on power ascension.		
	RO calculates dilution required to achieve ARO prior to 98.5% power.		
	PO sets up Main Turbine for power ascension to 98.5% at 10% per hour.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
pcm will have to be added via dilution. Note: Differential boron worth at 90% power for 6,000 EFPH is ~ -6.645 pcm/ppm. 108.6 pcm / -6.645 pcm/ppm = a boron concentration change of 16.3 ppm required. This can be calculated to be 1215 gallons of dilution total would be required in addition to the rods. (Calc found on Figure 101 of S2.RE-RA.ZZ-0012 Figures). Note: If dilution is used first, the turbine up power will not be initiated until after the RO reports the effect of the dilution is being observed in the RCS. Note: Auto or manual rod control is at the discretion of the CRS.			
	RO initiates dilution. (if required)		
	PO initiates power ascension to 98.5% at 10% per hour.		
	RO/PO monitor plant response to ensure power ascension is progressing as anticipated.		
	RO either announces expected and actual auto rod movement, or withdraws rods in manual with CRS concurrence to maintain Tavg on program.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Proceed to next event on direction from Lead Evaluator.			
2. 2C EDG Air start compressor leak/failure			
Simulator Operator: Insert RT-1 on direction from Lead Evaluator.			
MALF: AN0266 SER 266 FAILS- :J20 2C DIESEL GENERATOR URGENT FAILURE			
	PO announces OHA J-20, 2C DG URGENT TRBL, as unexpected.		
	PO refers to ARP and reports possible causes.		
	Crew dispatches operator to 2C EDG.		
Note: 2C EDG Air System is shown on drawing 205241 Sheet 6.			
Role Play: 2 minutes after being dispatched, report that when you arrived at 2C EDG control room, you could hear an air leak in the 2C EDG room. You acknowledged 2C EDG local alarm panel alarms A-4 and A-5, Air Receiver #1 Low Pressure and Air Receiver #2 Low Pressure respectively. In 2C EDG room you identified both EDG air start compressors running, and a			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
large air leak on the starting air supply line at the connection where the single supply line from the Moisture Separator splits to go to the two Air Receiver Tanks. Current Starting Air Tank pressures are both 176 psig and dropping. If asked, report the alarm setpoint for the low air pressure is 182 psig per the ARP.			
Note: If directed to isolate the air starting tanks inlet valves to isolate the leak, report back after 8 minutes that you have shut 21DA5C and 22DA5C (or the valves directed by CRS if different) to isolate the 23A and 23B Air Receiver Tanks respectively, and that both Air Start tanks are at 155 psig.			
Role Play: <u>IF</u> requested to inform CRS when air pressure reaches 160 psig, report back in 8 minutes that Air start pressure is 160 psig and lowering.			
Assuming a constant rate of pressure reduction, the Air Start tanks will reach their Tech Spec operability limit of 160 psig in 8 minutes from the initial field report.			
Note: CRS may direct starting 2C			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
EDG to ensure it remains operable. This would preclude having to declare the EDG inoperable, and extend the amount of time until the EDG would be in a 72 hour shutdown LCO (to allow a trouble shooting plan to be developed, obtain repair parts, etc). However, running an EDG unloaded for an extended period of time would require additional evaluation, and in any event the EDG will fail if it is attempted to be started.			
IF CRS orders PO to start 2C EDG from control console it will not reach proper speed, and stop shortly after the start attempt.			
IF CRS directs 2C EDG to be started locally, state you will review S2.OP-SO.DG-0003, 2C Diesel Generator Operation, ensure prerequisites are met, review the P&L's, do the pre-startup checks, and report back when you are ready to start 2C EDG.			
IF the CRS directs starting of 2C EDG without completing the step above, continue to next step.			
IF directed to locally start 2C EDG without performing normal start pre-req's, P&L's, etc., insert RT-10			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
to place the local start-stop switch to start, and report when 2C EDG trips 10 seconds later on over crank. (Local Alarm Panel window B-5 Failure to Start – Overcrank.)			
	CRS enters TSAS 3.8.1.1.b action b and determines the remaining 2 EDGs must be verified operable by performing S2.OP-ST-500-1, ELECTRICAL POWER SYSTEMS AC SOURCES ALIGNMENT, within 1 hour.		
Proceed to next event on direction from Lead Evaluator.			
3. 23 CCW pump w/failure of standby pump to start.			
Simulator Operator: Insert RT-3 on direction from Lead evaluator. MALFS: CC0172C 23 COMPONENT COOLING WATER PUMP TRIP RC007D 24 RC PUMP-#1 SEAL FAILS w/DEGRADATION Initial Value: 0.3 Severity: 0.75 Ramp: 5 minutes The 24 RCP high seal leakoff flow alarm will annunciate just before the 5 minute ramp is			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
finished.			
	RO reports trip of 23 CCW pump and standby pump did not auto start.		
	CRS directs RO to manually start 22 CCW pump.		
	RO manually starts 22 CCW pump and reports clearing of all alarms associated with the CCW pump trip.		
	CRS directs RO/PO to validate alarms received with ARP guidance.		
	RO/PO report that alarms received were consistent with low CCW system pressure, and that the restoration of system pressure was expected to clear all those alarms.		
	RO verifies RCP CCW cooled parameters trending to normal values.		
	CRS may enter S2.OP-AB.CC-0001, Component Cooling Abnormality, and S2.OP-AB.RCP-0001, Reactor Coolant Pump Abnormality, to verify actions previously performed have corrected condition.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Entry into these 2 AB's is not required because the action taken to restore CCW system pressure corrected the entry conditions for the AB's. Role Play: 2 minutes after being dispatched to check 23 CCW pump breaker, report overcurrent trip is present on 23 CCW pump. If dispatched to check CCW pumps, report they look normal. Note: Post scenario question may need to be asked for this Tech Spec call if it is not addressed prior to RCP malfunction. 4. 24 RCP seal degradation w/ failure Note: Seal inlet and outlet temps will lower, charging flow will rise, and RCDT level will rise (slightly), due to the seal leak.			
	CRS dispatches operators to investigate 23 CCW pump trip.		
	CRS enters TSAS 3.7.3 based on not having 2 operable CCW loops.		
	RO reports 24 RCP seal leakoff flow alarm current reading.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS enters S2.OP-AB.RCP-0001, Reactor Coolant Pump Abnormality.		
	CRS verifies entry condition of seal leakoff flow >5 gpm is met.		
	CRS directs initiation of CAS.		
	RO uses P-250 computer to check 24 RCP conditions.		
	CRS determines CCW water supply is not causing abnormal condition.		
	RO reports all RCP temperatures are normal and stable.		
Simulator Operator: MODIFY MALF RC007D from 0.75 to 1.0 with a 5 minute ramp on direction from Lead Evaluator	RO reports seal injection has not been lost.		
	RO reports seal leakoff is stable at 5.2 gpm.		
	CRS determines a unit shutdown should be performed.		
	RO reports seal leak has begun to rise towards 6 gpm.		
	CRS determines 24 RCP seal leakoff flow rising towards >6 gpm and will soon exceed 6 gpm CAS value, and goes to Attachment 2,		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: When fully ramped in, the MALF will cause 24 RCP seal leakoff to be >6 gpm. This is indicated by a console reading of "+OVER" on the hi range seal leakoff chart recorder.	Stopping Reactor Coolant Pumps.		
	CRS briefs Rx trip actions to be performed.		
	CRS identifies the requirement to shut 24CV104 3-5 minutes after pump shutdown.		
	RO trips the reactor.		
	RO stops 24 RCP, and crew recognizes 24CV104 must be shut in 3-5 minutes.		
	RO continues performing EOP-TRIP-1, Reactor Trip or Safety Injection, immediate actions from memory:		
	- Reports the Rx is tripped - Confirms Rx trip - Backs up the Main Turbine trip - Reports at least one 4KV vital bus energized. - Reports Safety Injection not actuated. - Reports Safety injection not required based on board indications.		
	CRS reads immediate action steps to RO and RO verifies performance.		
	PO receives CRS permission and throttles		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	AFW flow to no less than 22E4 lb/hr.		
	RO shuts 24CV104 3-5 minutes after 24 RCP is tripped.		
CT#1: Close 24CV104 3-5 minutes after tripping the RCP due to Seal Leakoff > 6 gpm. SAT_____ UNSAT_____			
Note: STA reports to the Control Room 10 minutes after being paged to report.	CRS transitions to EOP-TRIP-2, Reactor Trip Response.		
	RO makes page announcement "Unit 2 Rx trip."		
	PO reports total AFW flow is >22E4 lbm/hr.		
Simulator Operator: Ensure ET-3 is TRUE when 21 SGFP is tripped. This inserts the steam leak after a one minute delay.	PO stops 21 and 22 SGFPs.		
5. Steam leak in containment			
	PO lowers 23 AFW pump speed to minimum, and throttles 21-24AF21's to maintain AFW		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
6. MSLI failure	flow ~ 22E4 lbm/hr.		
	RO reports 3 RCPs running.		
	RO reports RCS Tavg is stable at or trending to 547° F.		
	RO reports unexpected OHAs for CFCU leak detection and Containment Sump pump start when they occur.		
	RO reports no indication of RCS leak.		
	Crew diagnoses steam leak in containment.		
	CRS orders a MSLI using AB.STM-1 CAS justification.		
	RO reports MSLI has failed on both trains.		
	CRS orders a Safety Injection.		
	RO initiates Safety Injection.		
	CRS returns to TRIP-1 based on TRIP-2 CAS of Safety Injection actuation.		
	CRS/RO verify immediate actions complete.		
	CRS dispatches operators to locally shut MSIVs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: See actions on next page for when containment pressure rises above 15 psig.	PO reports all available equipment started on SECs.		
	PO reports 21 and 22 AFW pumps running.		
	PO reports Table B valve groups in safeguards position.		
	RO reports 21/22CA330s are shut.		
	RO reports containment pressure is <15 psig.		
	RO reports no high steam flow conditions on 2RP4.		
	PO reports all 4KV vital buses energized.		
	RO reports correct control room and switchgear room ventilation alignment.		
	RO reports 2 CCW pumps in service.		
	RO reports ECCS flow for current RCS pressure as expected.		
	PO reports AFW flow >22E4 lbm/hr.		
	RO reports RCP status, and RCS temp not stable at or trending to 547° F due to steam leak, and MSLI has failed to shut any MSIV.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
7. Containment Spray pumps fail to start. Note: Containment Spray is directed to be started in TRIP-1 and in FRCE-1 when containment pressure reaches 15 psig. TRIP-1 Spray initiation (FRCE-1 initiation is on page 24)	RO reports both RTBs open.		
	RO reports both PZR PORVs shut and Block Valves open.		
	CRS refers/returns to TRIP-1 Step 11 and: - Ensures Containment Spray by initiating Phase B and Spray actuation. - RO inserts BOTH keys into CS actuation and turns at same time. - RO reports spray valve realignment. - Start 21 and 22 CS pumps - If SECs are not reset, RO blocks and resets 2A and 2C SECs. - RO inserts key and turns for 21 and 22 CS pumps and reports both CS pumps are running. - Initiate MSLI - RO reports MSLI failure. - Stop 21-24 RCPs - RO depresses stop PB for all RCPs. - Ensure valve groups in Table D in		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
CT #2 (CT-3) Manually start Containment Spray pumps prior to completion of step 3 of FRCE-1. SAT _____ UNSAT _____	safeguards position. - PO reports Table D valves position.		
	RO reports RCP status, and spray valves shut if RCPs in service.		
	PO reports all SG pressures dropping in an uncontrolled manner and MSLI has failed to shut any MSIV		
	CRS transitions to LOSC-1, Loss of Secondary Coolant.		
	RO reports MSLI failed to shut any MSIV, and dispatches operators to shut MSIVs if not previously performed.		
	PO reports all SG pressures dropping in an uncontrolled manner.		
	CRS transitions to LOSC-2, Multiple Steam Generator Depressurization.		
	PO ensures valves in Table A shut, and dispatches operator to shut any open valve if not previously performed.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Step 5 asks for cooldown rate, not if RCS has cooled down 100°F in last hour, which would be FRTS-1 (Thermal Shock) entry. Note: RCS C/D rate is ~ 120°F/hr.	PO reports 23 AFW pump not needed for SG level control, then trips and stops 23 AFW pump.		
	CRS dispatches operators to shut 23 AFW pump steam supply valves.		
	RO reports RCS cooldown rate is > 100°F per hour.		
	PO reduces AFW flow to each SG to no less than 1.0 E4 lbm/hr.		
CT#3 (CT-33) Reduce AFW flow to no less than 1.0E4 lbm/hr to each SG to minimize RCS cooldown prior to exiting FRCE-1. SAT _____ UNSAT _____			
	STA reports valid RED path on FRHS.		
	CRS transitions to FRHS-1, verifies it was operator action which caused the RED path, and returns to procedure in effect.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
FRCE-1 actions here			
	CRS directs closure of Phase A isolation valves not needed for recovery.		
	RO reports containment pressure >15 psig.		
	RO reports LOCA-5 not in effect.		
	RO starts or reports 21 and 22 CS in service.		
	RO reports 21 and 22CS2 are open.		
	RO reports 2CS14,16 and 17 are open.		
	RO stops RCPs if running.		
	RO reports all CFCUs running in slow speed with adequate SW flow.		
	RO reports MSLI valves all open, and dispatches operators to shut them if not previously performed.		
	PO lowers AFW flow to no less than 1.0E4 lbm/hr to each SG.		
	PO reports SG isolation valve status.		
	CRS returns to procedure in effect.		
Simulator Operator: Once AFW flow has been lowered to SGs, and AFTER FRCE-1 has been			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
performed, and BEFORE reaching step 8 of LOSC-2, THEN remove MALF from the MSIV the CRS directed to be closed first that is NOT 21 MSIV.			
Terminate scenario upon transition back to LOSC-1.	PO reports rising pressure in affected SG.		
	CRS uses CAS action of LOSC-2 of rising pressure in any SG to return to LOSC-1.		

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-IO.ZZ-0004, Power Operation
- F. S2.OP-SO.DG-0003, 2C EDG Operation
- G. S2.OP-AR.DG-0003, 2C Diesel Generator Alarm Response
- H. Dwg 205241, Sheet 6, Diesel Engine Auxiliaries
- I. S2.OP-AB.CC-0001, Component Cooling System Abnormality
- J. S2.OP-AB.RCP-0001, Reactor Coolant Pump Abnormality
- K. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- L. 2-EOP-TRIP-2, Reactor Trip Response
- M. 2-EOP-LOSC-1, Loss of Secondary Coolant
- N. 2-EOP-LOSC-2, Multiple Steam Generator Depressurization
- O. 2-EOP-FRCE-1, Response to Excessive Containment Pressure

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 90 RCS BORON: 895 MWe 1080

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Core Burnup: 6,000 EFPH

Control Bank D is at 203 steps.

Reactivity plan for power ascension to 98.5% is to perform dilution concurrent with rod withdrawal to establish ARO prior to reaching 98.5%.

Calorimetric required prior to exceeding 90% power has been performed SAT.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

S2.OP-IO.ZZ-0004, Power Operation, in progress and complete up to Step 5.1.26.

Power was reduced 2 days ago to remove 21 and 22 Heater Drain Pumps from service due to possible damage from spraying water. Pumps have been inspected and returned to service.

ABNORMAL PLANT CONFIGURATIONS:

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

SECONDARY:

Heating steam is aligned to Unit 1.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
___ With Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct if applicable.
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 13-01 NRC-ESG-1

REVIEWER: P Williams

Initials	Qualitative Attributes
----------	------------------------

- | | | |
|----|-----|---|
| PW | 1. | The scenario has clearly stated objectives in the scenario. |
| PW | 2. | The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events. |
| PW | 3. | The scenario consists mostly of related events. |
| PW | 4. | Each event description consists of: <ul style="list-style-type: none"> • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point |
| PW | 5. | No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event. |
| PW | 6. | The events are valid with regard to physics and thermodynamics. |
| PW | 7. | Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives. |
| PW | 8. | The simulator modeling is not altered. |
| PW | 9. | All crew competencies can be evaluated. |
| PW | 10. | The scenario has been validated. |
| PW | 11. | If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario. |
| PW | 12. | ESG-PSA Evaluation Form is completed for the scenario at the applicable facility. |

ATTACHMENT 4 **SIMULATOR SCENARIO REVIEW CHECKLIST**

Note: The quantitative attribute target ranges that are specified on the form are not absolute limitations; some scenarios may be an excellent evaluation tool, but may not fit within the ranges. A scenario that does not fit into these ranges shall be evaluated to ensure that the level of difficulty is appropriate. (ES-301 Section D.5.d)

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)
GG	7	Total malfunctions inserted: 5-8
GG	3	Malfunctions that occur after EOP entry: 1-2
GG	3	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	1	EOP contingencies requiring substantive actions: 0-2
GG	3	Crew Critical Tasks: 2-3

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

13-01 NRC ESG-1

CT#1: (AB.RCP) Close 24CV104 3-5 minutes after tripping the RCP due to Seal Leakoff > 6 gpm.

BASIS: Closing the Seal Leakoff Isolation Valve 24CV104 prevents unfiltered, high temperature RCS into the #2 RCP seal, which raises the probability of #2 seal failure. The 3 minute time frame is based on expected coastdown time of the RCP since a static condition is more favorable for the #2 seal to perform its backup function to the #1 seal. The 5 minute limitation minimizes RCS leakage from the damaged seal to maintain RCS inventory. Failure to isolate within 5 minutes raises the possibility of a #2 seal failure and subsequent LOCA.

CT#2 (CT-3) Manually start Containment Spray pumps prior to completion of step 3 of FRCE-1.

BASIS: Failure to manually actuate the minimum required complement of containment cooling equipment under the postulated conditions demonstrates the inability of the crew to "recognize a failure or an incorrect automatic actuation of an ESF system or component."

CT#3 (CT-33) Reduce AFW flow to no less than 1.0E4 lbm/hr to each SG to minimize RCS cooldown prior to exiting FRCE-1.

BASIS: Failure to perform the critical task increases the challenges to the subcriticality and the containment CSFs beyond those irreparably introduced by the postulated plant conditions.

EVENTS LEADING TO CORE DAMAGE

<u>Y/N</u>	<u>Event</u>
N	TRANSIENTS with PCS Unavailable
N	Steam Generator Tube Rupture
N	Loss of Offsite Power
N	Loss of Switchgear and Pen Area Ventilation
N	LOCA

<u>Y/N</u>	<u>Event</u>
N	Loss of Service Water
Y	Loss of CCW
N	Loss of Control Air
N	Station Black Out

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCYY/N COMPONENT, SYSTEM, OR TRAIN

N	Containment Sump Strainers
N	SSWS Valves to Turbine Generator Area
N	RHR Suction Line valves from Hot Leg
N	CVCS Letdown line Control and Isolation Valves

Y/N COMPONENT, SYSTEM, OR TRAIN

N	Gas Turbine
Y	Any Diesel Generator
N	Auxiliary Feed Pump
N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGEY/N OPERATOR ACTION

N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.